

Lab Sheet # L13

Instructions:

- Submit *.sql file, named as Lab13_6XXXXXX.sql to LAs or instructors prior to My Courses.
- If students cannot finish within the class time, students can submit until the beginning of the next class.

Self-study:

<https://www.mysqltutorial.org/mysql-stored-procedure/mysql-transactions/>

<https://www.mysqltutorial.org/mysql-basics/mysql-table-locking/>

<https://www.mysqltutorial.org/mysql-stored-procedure/sql-cursor-in-stored-procedures/>

Today's Task

0. Database Setup:

Import the database by executing the **classicmodels_db.sql** file. (Same as Lab2)

For the INSERT / UPDATE task, you can put any values.

	Easy Task: Transaction + Table Locking + Cursor
1.	Start a transaction to insert a new customer into the customers table. Then commit the transaction.
2.	Start a transaction to update the creditLimit of customer 103. Rollback the transaction.
3.	Start a transaction to insert a new order into the orders table. Commit the transaction.
4.	Start a transaction to update the status of an order. Commit the transaction.
5.	Start a transaction to delete a payment record from payments . Rollback the transaction.
6.	Start a transaction to update products.quantityInStock . Commit the transaction.
7.	<u>Disable autocommit mode</u> , start a transaction to insert a new product, then commit.
8.	Write SQL statements to: • Lock the table products using WRITE mode • Update the value of quantityInStock for a product • Unlock the table
9.	Create a stored procedure named: list_employee_emails The procedure must: • Use a <u>cursor</u> • Retrieve all employee emails from the employees table

	Concatenate them into a single text variable.
10.	Create a stored procedure named: list_customer_names The procedure must: - Use a <u>cursor</u> - Retrieve all customer names - Combine them into a single string separated by commas.
	Medium Task: transaction logic + cursor iteration
11.	Write SQL statements using a transaction to: - Insert a new order into orders - Insert its corresponding record into orderdetails Commit the transaction only after both statements succeed.
12.	Start a transaction to reduce products.quantityInStock when an order is placed. Rollback the transaction if the stock becomes negative.
13.	Write a transaction that: - Updates a customer's creditLimit - Inserts a new payment record into payments Commit the transaction.
14.	Start a transaction that updates an employee's officeCode. Rollback the transaction.
15.	Create a stored procedure that uses a cursor to: - iterate through all rows in orders - count the number of orders for each customer.
16.	Create a stored procedure using a cursor to calculate the total payment amount for each customer.
17.	Create a stored procedure using a cursor to: retrieve all products with quantityInStock < 100.
	Hard Task: realistic transactional workflow
18.	Write a transaction to transfer credit from one customer to another: - reduce creditLimit from customer A - increase creditLimit of customer B Commit the transaction only if both updates succeed.
19.	Write a transaction that: - Inserts a new order into orders - Inserts multiple records into orderdetails Rollback the transaction if any insert fails.
20.	Write a transaction to update stock for all products included in an order. If any product does not have enough stock: Rollback the entire transaction.